# ALGEBRAIZING RIG-ETALE MORPHISMS OVER A FORMAL COMPLETION

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-algebraize-morphism-rig-etale; the proof environment immediately following it is its own proof

*Editorial note.* The label lemma-algebraize-morphism-rig-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- the /T completion functor: a morphism of schemes g : V to X induces g_{/T} : V_{/g^{-1}T} to X_{/T} of formal algebraic spaces, functorially in schemes over X (section-completion-functor)
- the fully faithful lemma of the same chapter (lemma-fully-faithful-etale-over-complement): morphisms between formal completions of etale-over-the-complement schemes come from unique morphisms of the schemes, and etale arrows algebraize

*Source reference.* Original reference 0AKJ: sources/stacks-restricted/pinned-original-source.tex, source lines 2065–2065. Prerequisite pointer invoked at proof line(s) 6450; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original section 0GDB: sources/stacks-restricted/pinned-original-source.tex, source lines 5953–5953. Prerequisite pointer invoked at proof line(s) 6448; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Assume we have*

(1) *Noetherian affine schemes $X$, $X'$, and $Y$,*
(2) *a closed subset $T \subset |X|$,*
(3) *a morphism $f : X' \to X$ locally of finite type and étale over $X \setminus T$,*
(4) *a morphism $h : Y \to X$,*
(5) *a morphism $\alpha : Y_{/T} \to X'_{/T}$ over $X_{/T}$ (see proof for notation).*

*Then there exists an étale morphism $b : Y' \to Y$ of affine schemes which induces an isomorphism $b_{/T} : Y'_{/T} \to Y_{/T}$ and a morphism $a : Y' \to X'$ over $X$ such that $\alpha = a_{/T} \circ b_{/T}^{-1}$.*

*Date*: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* The notation using the subscript $_{/T}$ in the statement refers to the construction which to a morphism of schemes $g : V \to X$ associates the morphism $g_{/T} : V_{/g^{-1}T} \to X_{/T}$ of formal algebraic spaces; it is a functor from the category of schemes over $X$ to the category of formal algebraic spaces over $X_{/T}$, see Section 0GDB. Having said this, the lemma is just a reformulation of Lemma 0AKJ. □